

Kavya Shah

kashah@ucsd.edu | linkedin.com/in/kavyashah3008 | +1 (858) 900-8447

Education

University of California, San Diego

Expected 2027

BS in Mathematics and Computer Science, Minor in Data Science

GPA: 3.89

- **Relevant Coursework:** Machine Learning, Data Visualization, Database Systems, Probability and Statistics, Software Engineering, Theoretical Foundations of Data Science, Data Structures, Object-Oriented Programming in Java, Software Tools and Techniques Lab, Linear Algebra, Vector Calculus, Discrete Mathematics, Calculus I/II/III.

Skills

- **Languages:** Python, Java, C++, C, SQL, JavaScript, R, MATLAB.
- **Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph, Redis, OpenCV, Pandas, React, Flask.
- **Concepts:** Agentic applications, Multi-agent systems, Deep Agents, RAG, Fine-tuning, Cloud AI Deployment.

Professional Experience

AI Consultant, Pear.care

Sept 2025 – Present

Remote

- Led development of **Deep Agents** orchestrating medical workflows (symptoms→ICD→CPT→doctors) through contextual routing.
- Integrated **MCP (Model Context Protocol)** to enable agents to securely access standardized research, calculator, and audit tools.
- Built scalable **RAG-based AWS pipelines** with multi-tenant security, ethical safeguards, and task scheduling for clinical AI.
- Developed **semantic search pipelines** using embeddings and LLMs to interpret medical intent and retrieve evidence.

Database Engineer, San Diego Supercomputer Center (SDSC)

Aug 2025 – Present

San Diego, CA

- Engineered secure **data migration pipelines** extracting congenital heart disease datasets from **AWS DynamoDB** using **Boto3**.
- Transformed semi-structured clinical and MRI metadata into standardized **JSON schemas** and migrated data into **PostgreSQL**.
- Built reproducible **PostgreSQL dumps** and **Dockerized pipelines** enabling scalable analytics and multi-center research access.

Project Manager, Skin Lesion Detection (CSE Society @UCSD)

2024 – Present

San Diego, CA

- Led design of a **melanoma detection model** using **ResNet-50 CNNs**, transfer learning, and cross-validation techniques.
- Developed **image preprocessing pipelines** with resizing, normalization, and augmentation to improve deep model robustness.
- Implemented **TensorFlow** and **PyTorch** workflows for model quantization, compression, and efficient on-device inference.
- Achieved 91.3% accuracy, 0.89 ROC-AUC, and presented research findings at **SAIRS 2025 Conference**.

Research Assistant - Bioinformatics & Data Analytics, UC San Diego Health

2025 – Present

San Diego, CA

- Designed **ML pipelines for scRNA-seq**, integrated with ATAC-seq and RNA velocity for multimodal immune profiling.
- Built Scanpy & scVI pipelines integrating scRNA-seq & ATAC-seq; deployed reproducible **multimodal workflows**.
- Discovered **tumor-specific immune cell clusters** post-immunotherapy, providing potential biomarkers for treatment response.

Machine Learning Intern, F(X) Data Labs

2024

Ahmedabad, India

- Analyzed IoT sensor data and customer logs to **detect predictive failure patterns** for proactive equipment maintenance.
- Developed and **fine-tuned LSTM models** with advanced feature engineering to capture temporal dependencies in sensor readings.
- Applied **clustering and anomaly detection** to identify hidden failure modes and optimize maintenance scheduling strategies.

Projects

Travelog – San Diego Supercomputer Center (*Java, C++, React, Node.js, Flask, MongoDB, AWS, Next.js*)

2025

- Engineered AI-assisted travel planning features using React, Flask APIs, and integrated user preference learning models.
- Deployed AWS CI/CD pipelines that reduced UI-API deployment bugs by 30% and improved production stability overall.

Predicting Credit Default: Age × Behavior (*Python, scikit-learn, SHAP, LangChain, Pandas/NumPy*)

2025

- Created agentic workflow decomposing credit risk tasks into feature engineering, validation, and explainability subtasks.
- Fine-tuned deep learning models with SHAP explainability on 150K+ U.S. credit profiles for transparency.
- Improved AUC by 12% through interaction features; LangChain validation and explainability showed 18% higher variance explained.

AI-Powered Code Review Chatbot (*Python, GPT API, Flask, LangGraph, TypeScript*)

2025

- Developed GPT+AST chatbot with LangGraph orchestration; Redis enabled contextual memory for accurate pull request feedback.
- Integrated TypeScript-based frontend with Flask backend, enabling real-time review feedback and interactive developer dashboards.

Face Recognition System with OpenCV (*Python, OpenCV, Haar Cascades, Cosine Similarity*)

2024

- Implemented secure recognition system with GUI login interface and identity embeddings for efficient authentication.
- Achieved 95.7% accuracy on 10K images, reducing false positives by 30% consistently.
- Utilized OpenCV Haar cascades for real-time face detection, paired with cosine similarity for robust matching.

Edge Detection with Hough Transform (*Python, OpenCV, NumPy, Go*)

2024

- Designed lane detection pipeline using Canny edges and Hough Transform, optimized for real-world road conditions.
- Rebuilt inference module in Go for real-time performance, achieving sub-50ms latency on highway video streams.